

KYLE J. UTLEY

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LinkedIn: linkedin.com/in/kyle-utley

Mechanical Engineering Student | Manufacturing & Process Engineering

EDUCATION

The University of Alabama

Expected Graduation May 2028

Tuscaloosa, AL — Bachelor Of Science: Mechanical Engineering Major, Automotive Minor

- Cumulative GPA: 3.67 | Member, University of Alabama Honors College | Accelerated STEM-to-MBA Program

RELEVANT EXPERIENCE

EQUIPMENT ENGINEERING CO-OP – LINE 1

Aug 2024 - Dec 2024 | Jan 2025 - Apr 2025

Honda Motors | Lincoln, AL

- Modeled the new rear suspension lift-assist in CATIA, applying DFM principles, then fabricated the frame in-house with a plasma cutter, lathe, and mill, joined via MIG welding; consolidated pneumatic controls, cut unit weight by 7.0 lb, and added rubber arm limiters to ease rotational contact and reduce associate strain.
- Achieved zero unplanned downtime since installation and positive associate feedback, with a more serviceable design that simplified ongoing maintenance and troubleshooting.
- Diagnosed a recurring part-presence sensor jam as a mechanical design flaw, iterating actuator flag material (metal, nylon, Delrin) to eliminate wear; redesigned switches have had zero jams since installation, preventing associates from resorting to manually handling batteries and risking drops, vehicle damage, and injury.
- Manufactured in-house replacement parts for worn equipment components, avoiding costly expedited outsourcing to external machine shops and meaningfully reducing part-replacement spend.

SUSPENSION TEAM MEMBER

Jan 2024 - Present

Crimson Racing – Formula SAE, The University of Alabama | Tuscaloosa, AL

- Designed a new rear bellcrank in SolidWorks, applying FEA to validate structural performance against defined cornering load cases before machining the final part from 7075-T6 aluminum to minimize mass without compromising strength.
- Bellcrank held up across every drive day and competition run of the season without a single failure, validating both the design and the manufacturing process under real operating loads.
- Designed and waterjet-manufactured a precision-slotted tone ring and magnetic sensor mounting bracket to develop a new wheel-speed detection method, prioritizing manufacturability and installation precision at the wheel hub.
- Validated the new detection method across multiple drive days, cross-referencing calculated wheel speed against the team's existing sensing system to confirm data accuracy, successfully proving the method for production use across the team's combustion and electric race cars.

ADDITIONAL EXPERIENCE

STUDENT ASSISTANT

July 2026 - Present

Manderson Graduate School of Business, The University of Alabama | Tuscaloosa, AL

- Provide cross-functional support across reception, records management, event logistics, and communications, serving as a primary point of contact for visitors, students, and faculty.

TECHNICAL SKILLS

Shop & Fabrication

- CNC Machining | 3D Printing | Waterjet | Plasma Cutter
- MIG Welding | Lathe | Mill | Press | Grinder | Bandsaw

CAD & Design

- SolidWorks | CATIA | FEA Analysis | Fusion 360
- Root Cause Analysis | DFMEA | DFM | Sensor Integration

INVOLVEMENT AND HONORS

- Member, Alabama AI's Pals Mentorship Program
- Member, American Society of Mechanical Engineers
- University of Alabama Dean's List | Fall 2025
- University of Alabama President's List | Spring 2024